THE COMPLEMENT OF THE LOCUS WHERE A SECTION IS INVERTIBLE HAS CODIMENSION ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Extension of reflexive rank one modules over a complement of codimension at least two. Lemma lemma-reflexive-S2-extend, used at proof lines 7532-7535. Promotes the isomorphism across the complement.
- Serre's criterion for normality. Properties, Lemma properties-lemma-criterion-normal, used at proof lines 7532-7535. Supplies the hypotheses of the extension criterion.

Source reference. Original lemma 0EBJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17924–17924. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0EBN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17928–17928. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0345: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4080–4080. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Assumptions and notation as in Lemma 0EBN. If s is nonzero, then every irreducible component of $X \setminus U$ has codimension 1 in X.*

Proof. Let $\xi \in X$ be a generic point of an irreducible component Z of $X \setminus U$. After replacing X by an open neighbourhood of ξ we may assume that $Z = X \setminus U$ is irreducible. Since $s : \mathcal{O}_U \to \mathcal{F}|_U$ is an isomorphism, if the codimension of Z in X is ≥ 2, then $s : \mathcal{O}_X \to \mathcal{F}$ is an isomorphism by Lemma 0EBJ and Serre's criterion Properties, Lemma 0345. This would mean that $Z = \emptyset$, a contradiction. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.